

IMPROVING FARM PRODUCTIVITY WITH ML-BASED PRECISION TECHNIQUE

An Engineering Project in Community Service

Phase – I Report

Submitted by

- | | |
|----------------------|------------------------|
| 1. 23BAI10014 | ROSHMIK AGRAWAL |
| 2. 23BAI10514 | DIVYANSH KUMAR |
| 3. 23BAI10866 | MOHD BELAL |
| 4. 23BCE10220 | AMOGH AGARWAL |
| 5. 23BCE10279 | AAYUSH SINGH |
| 6. 23MIP10136 | SURENDRA MEWADA |

in partial fulfillment of the requirements for the degree of

Bachelor of Technology



VIT[®]
BHOPAL
www.vitbhopal.ac.in

VIT Bhopal University
Bhopal
Madhya Pradesh

NOVEMBER 2025



Bonafide Certificate

Certified that this project report titled **IMPROVING FARM PRODUCTIVITY WITH ML-BASED PRECISION TECHNIQUE** is the bonafide work of **23BAI10014 ROSMIK AGRAWAL, 23BAI10514 DIVYANSH KUMAR, 23BAI10866 MOHD BELAL, 23BCE10220 AMOGH AGARWAL, 23BCE10279 AAYUSH SINGH, 23MIP10136 SURENDRA MEWADA** who carried out the project work under my supervision.

This project report (Phase I) is submitted for the Project Viva-Voce examination held on 26/11/2025

Supervisor

Comments & Signature (Reviewer 1)

Comments & Signature (Reviewer 2)



VIT[®]
BHOPAL
www.vitbhopal.ac.in

Declaration of Originality

We, hereby declare that this report entitled **IMPROVING FARM PRODUCTIVITY WITH ML-BASED PRECISION TECHNIQUE** represents our original work carried out for the EPICS project as a student of VIT Bhopal University and, to the best of our knowledge, it contains no material previously published or written by another person, nor any material presented for the award of any other degree or diploma of VIT Bhopal University or any other institution. Works of other authors cited in this report have been duly acknowledged under the section "References".

Date- 26/11/2025

Reg No & Name

23BAI10014 ROSMIK AGRAWAL

23BAI10514 DIVYANSH KUMAR

23BAI10866 MOHD BELAL

23BCE10220 AMOGH AGARWAL

23BCE10279 AAYUSH SINGH

23MIP10136 SURENDRA MEWADA

Acknowledgement

First and foremost, we would like to thank the Lord Almighty for His presence and immense blessings throughout the project work.

We wish to express my heartfelt gratitude to **Dr. Pushpinder Singh Patheja**, Head of the Department, School of Computer Science and Engineering for much of his valuable support and encouragement in carrying out this work.

We would like to thank our Programme Chair **Dr. Pradeep Kumar Mishra (SCAI)**, **Dr. Vikas Panthi (SCOPE)**, for giving us this valuable opportunity to present this project.

We would like to thank our internal guide **Dr. Adnan Abbassi**, for continually guiding and actively participating in our project, giving valuable suggestions to complete the project work.

We would like to thank all the technical and teaching staff of the School of Computer Science and Engineering, who extended directly or indirectly all support.

Last, but not least, we are deeply indebted to our parents who have been the greatest support while we worked day and night for the project to make it a success.

Abstract

Agriculture remains a vital pillar of the Indian economy, yet farmers continue to face challenges such as poor crop selection, unpredictable climate conditions, and reliance on traditional farming practices. These issues often lead to reduced yields, increased financial risks, and unsustainable farming methods. To address these concerns, this project proposes a Machine Learning (ML)-based precision technique designed to enhance farm productivity through intelligent and data-driven crop recommendations.

The system utilizes soil nutrient values, climatic conditions, and geographical factors to build predictive models that identify the most suitable crops for a given environment. By integrating historical agricultural data with modern ML algorithms, the solution delivers accurate, context-aware recommendations that can significantly improve yield outcomes. The approach emphasizes sustainability, resource efficiency, and ease of use, making it practical for real-world agricultural applications.

This ML-based precision farming solution aims to empower farmers with scientific insights, reduce economic losses caused by uninformed decision-making, and promote long-term agricultural sustainability. The results of this project demonstrate the transformative potential of technology-driven tools in modernizing Indian agriculture and improving farmer livelihoods.

Index

Sl. No.	Topic	Page No.
1.	Introduction	1
1.1.	Motivation	2
1.2.	Objective	3
2.	Existing Work	4
2.1.	Soil Classification using Machine Learning Methods and Crop Suggestion Based on Soil Series, 2018:	
2.2.	Intelligent Crop Recommendation System using Machine Learning, 2021:	
2.3.	Crop Prediction Method to Maximize Crop Yield Using Machine Learning, 2020	
2.4.	Crop Yield Prediction Using Machine Learning: A Pragmatic Approach	
3.	Topic of the work	7
A.	System Design / Architecture	
B.	Working Principle	
C.	Results and Discussion	
D.	Individual Contribution by members	
4.	Conclusion	22
5.	Reference	23
6.	Biodata with Picture	25

1. INTRODUCTION

Agriculture plays a fundamental role in sustaining human civilization. In India, agriculture not only ensures food security but also supports rural livelihoods. However, modern agriculture faces severe challenges including:

- Climate change and irregular rainfall
- Soil nutrient depletion
- Increasing input costs
- Limited scientific advisory access
- Lack of data-driven farming practices

Crop selection is one of the most critical decisions in farming. Choosing an unsuitable crop for a particular soil and climatic condition can result in yield reduction, resource wastage, and economic loss.

Traditional decision-making in agriculture is often based on:

- Historical practices
- Neighboring farm patterns
- Local intuition
- Trial-and-error methods

These approaches are no longer sufficient in the era of climate variability.

Machine Learning provides a powerful framework for analyzing multi-dimensional agricultural data. By processing soil nutrients, weather parameters, and geographical inputs, ML models can generate precise and personalized recommendations.

This project enhances existing ML-based systems by integrating **geo-spatial intelligence using GPS and weather APIs**, transforming a static system into a dynamic precision agriculture platform.

1.1 Motivation

The motivation for this project emerged from observing real-world agricultural inefficiencies:

1. Farmers often lack access to region-specific advisory systems.
2. Environmental data is manually entered, leading to inaccuracies.
3. Yield predictions are not dynamically updated.
4. Many systems are research prototypes and not deployable.

Additionally, during the study of existing literature, it was identified that:

- Most research considers either soil parameters or climatic conditions independently.
- Very few systems integrate geographical location automatically.
- Real-time data integration is rarely implemented.

We aimed to bridge this gap by:

- Automating environmental data retrieval
- Integrating geo-location intelligence
- Building a deployable web-based system
- Ensuring high model accuracy and interpretability

Our ultimate goal is to empower farmers with a **scientifically backed digital advisory system**.

1.2 Objective

1. Develop a highly accurate crop recommendation model.
2. Develop a regression-based yield prediction system.
3. Integrate GPS-based automatic location detection.
4. Fetch environmental parameters dynamically via API.
5. Reduce manual dependency in agricultural decision-making.
6. Improve sustainability and productivity.
7. Deploy an accessible web-based system.
8. Provide interpretable and explainable predictions.

2. Existing Work / Literature Review

2.1 Soil Classification using Machine Learning Methods and Crop Suggestion Based on Soil Series, 2018:

Author: Sk Al Zaminur Rahman Kaushik Chandra Mitra S.M. Mohidul Islam

Objective: This study highlights how machine learning can help farmers identify soil types and recommend the best crops for their land. By analyzing factors like soil texture, pH level, organic matter, and nutrient content, the model predicts soil types and suggests crops that are most likely to thrive. This approach aims to improve crop yields by aligning farming decisions with the specific characteristics of the soil.

The research tested several machine learning algorithms, including Random Forest, Support Vector Machines (SVM), and k-Nearest Neighbors (k-NN). Among these, the Random Forest algorithm stood out, achieving 92% accuracy in classifying soil types. The analysis found that soil texture and pH level were the most critical factors for accurate predictions, making the model highly reliable for practical use.

Integrating soil analysis with crop recommendations can revolutionize farming by helping farmers make smarter, more sustainable decisions. By tailoring crop selection to soil conditions, this approach maximizes yields, reduces resource waste, and promotes efficient agricultural practices. This tool is a step toward improving productivity and supporting farmers in managing their land effectively.

2.2 Intelligent Crop Recommendation System using Machine Learning, 2021:

Authors: Priyadharshini A, Swapneel Chakraborty, Aayush Kumar, Omen Rajendra Pooniwala

Objectives: This article shows how machine learning can be used to help farmers choose the best crops. The aim is to make farming better and lessen money problems from bad crop choices. It aims to provide data for the farmer to understand the soil, seasons, and the land he or she is working on much better.

The article talks about a smart system encouraging farmers to cultivate the proper crops through machine learning. It is, in fact, an important part of the Indian economy because it comprises nearly 50% of the people. In case, inappropriate crops are selected, then low yields and financial issues may occur. These issues get worse due to ignorance of the kind of soil, weather, and geography. This literature explains how researchers would collect heavy data, including crop yield, soil nutrient, cultivation costs, weather condition, and crop prices, gathered from credible sources such as government websites and Kaggle. In managing data inconsistencies, they substituted missing values with -1 in order to have smoother predictions. To develop their system, they used multiple machine learning models, including linear

regression, neural networks, K-Nearest Neighbors (KNN), and decision trees. Their system can be categorized into three sections: profit analysis, crop recommendations, and sustainability predictions.

The neural network got an accuracy of 89.88%, linear regression was at 88.26%, and KNN with cross-validation was at 88%. The system is planned to have real uses, like a mobile app and a web interface, to give farmers personalized crop advice, helping millions. The writing demonstrates how technology, especially machine learning, can greatly enhance farming practices. It helps farmers make better choices and promotes progress in sustainable farming.

2.3 Crop Prediction Method to Maximize Crop Yield Using Machine Learning, 2020

Authors: Neelam Singh, Deeksha Pant, Devesh Pratap Singh, Bhaskar Pant

Objective: This study shows how machine learning can help farmers choose crops that suit their region, improve yields, and address agricultural challenges across India. For example, water-saving crops like wheat and mustard are ideal for Punjab and Haryana, while rice and pulses work well in monsoon-dependent states like Tamil Nadu. Drought-resistant crops like cotton and sorghum are recommended for Maharashtra and Gujarat.

Different machine learning models tackle specific challenges. K-Nearest Neighbors (KNN) predicts yields in states with varied climates, while Naïve Bayes helps regions with unpredictable rainfall like Madhya Pradesh. Random Forest handles complex factors like drought in Maharashtra, and K-Star works well in tough terrains like Himachal Pradesh. Future advancements could include real-time weather and soil monitoring, supported by mobile apps to provide farmers with instant advice. These innovations can transform Indian agriculture, boosting productivity and food security.

2.4 Crop Yield Prediction Using Machine Learning: A Pragmatic Approach

Authors: Rajswhee Surana, Ritu Khandelwal

Objective: To develop a machine learning model for accurate crop yield prediction to help farmers improve agricultural productivity. By evaluating algorithms like Random Forest, Adaboost, Gradient Boost, and SVM, the research aims to identify the key environmental factors influencing crop yields and determine the most accurate prediction method. The goal is to provide farmers with reliable insights for optimal crop selection, enhancing yield and resource utilization in a changing agricultural landscape.

This research uses machine learning to help farmers predict crop yields more accurately, improving productivity and reducing risks. It evaluates algorithms like Random Forest, Adaboost, Gradient Boost, and SVM, focusing on factors like soil pH, temperature, rainfall,

and humidity. A dataset of 2,201 samples was split for training (80%) and testing (20%). Random Forest proved most accurate.

The study aims to provide farmers with practical insights for better crop choices and efficient resource use, boosting yields and supporting smarter farming in a changing agricultural landscape.

3. Topic of the work

A. System Design / Architecture

i. Goal

The goal of this project is to help farmers make informed decisions on which crops to plant and predict the potential yield based on soil and environmental factors. The project involves two key processes: predicting the right crop to grow based on soil conditions (Crop Prediction) and forecasting the crop yield (Yield Prediction).

ii. Functional Modules Design and Analysis

The system has two main parts:

- **Crop Prediction:** Based on factors like soil nutrients and weather, it recommends the most suitable crop.
- **Yield Prediction:** It predicts the amount of crop yield based on factors like rainfall, temperature, and the use of pesticides.

iii. Methodology of Crop Prediction

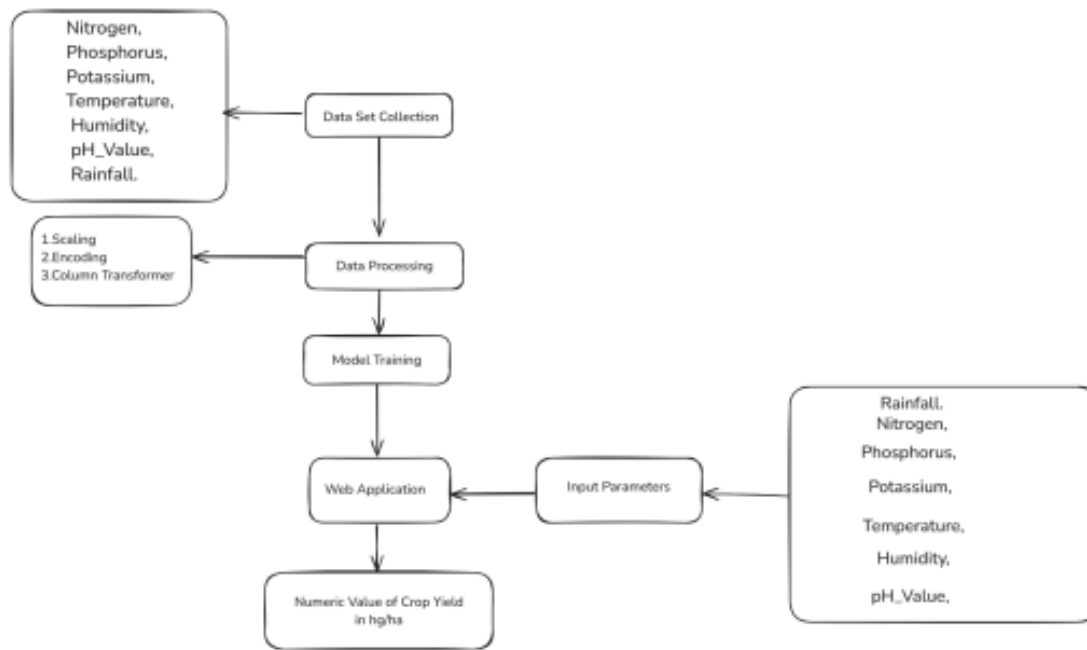
➤ Input Data

To make a recommendation, the model uses seven pieces of information:

- **Nitrogen:** Amount of nitrogen in the soil.
- **Phosphorus:** Amount of phosphorus in the soil.
- **Potassium:** Amount of potassium in the soil.
- **Temperature:** Weather temperature (°C).
- **Humidity:** Amount of moisture in the air.
- **pH Value:** The acidity or alkalinity of the soil.
- **Rainfall:** Total yearly rainfall (in millimeters).

➤ Target Output:

The model recommends a crop, such as "Wheat," "Rice," etc., based on these inputs.



➤ **Data Preprocessing**

The data undergoes a few simple steps to ensure it's ready for the model:

- **Cleaning Column Names:** Removes extra space to avoid errors in the column names.
- **Verify Dataset:** Ensures that all the expected data is available (like Nitrogen, Phosphorus, etc.). If not, it raises an error.
- **Label Encoding for Crop:** Converts crop names (like "Wheat") into numbers that the model can understand, which makes the target variable machine-readable.

➤ **Splitting the Dataset**

The dataset is split into two parts:

- **Training Data (80%):** Used to teach the model how to make predictions.
- **Testing Data (20%):** Used to test how well the model works on new data.

➤ **Algorithm Used: Random Forest Classifier**

This method uses several decision trees to predict the crop. Each tree makes its prediction, and the final recommendation is decided by the majority of vote. The more trees, the more accurate the result. The algorithm uses:

- **n_estimators=100:** 100 decision trees are created for better prediction.
- **random_state=42:** Ensures the results are consistent across different runs.

➤ **Model Training**

The model learns by using the training data and is adjusted to make accurate predictions about which crop to recommend.

➤ **Model Evaluation**

After training, the model is tested on new data to check its performance. The model's accuracy is calculated by comparing its predictions to the actual values in the test dataset.

Model Accuracy: 99.32%.

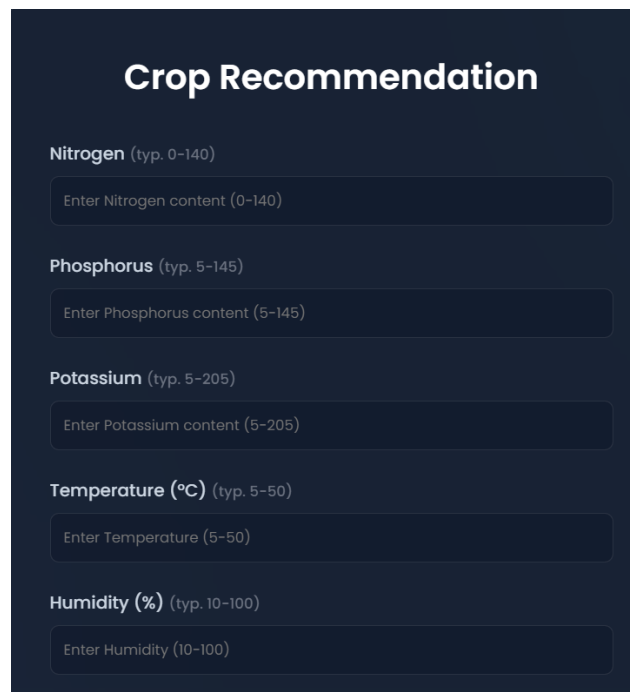
➤ **Model Saving**

Once trained, the model is saved in a file (e.g., crop_model.pkl, pkl - pickle file) so it can be used later for predictions. The LabelEncoder is also saved as label_encoder.pkl for consistent use of crop labels

iv. User Interface Designs

The web app is simple to use. For both crop and yield predictions, users input relevant data (like rainfall, temperature, etc.), and the system displays the results on a dashboard.

- **Input Forms:** Easy-to-use forms for entering data.



The image shows a dark-themed web application interface titled "Crop Recommendation". It features five input fields, each with a label and a range in parentheses: "Nitrogen (typ. 0-140)", "Phosphorus (typ. 5-145)", "Potassium (typ. 5-205)", "Temperature (°C) (typ. 5-50)", and "Humidity (%) (typ. 10-100)". Each label is followed by a text input box containing a placeholder text that matches the label's range (e.g., "Enter Nitrogen content (0-140)").

Crop Yield Prediction

Crop (e.g. Rice, Wheat)

Crop Year (e.g. 2023)

Annual Rainfall (mm/year)

Pesticide (tonnes)

Fertilizer (tonnes)

Area (Hectares)

State (e.g. Assam)

Season (e.g. Kharif)

- **Output Dashboard:** Shows the crop recommendation or yield production

Nitrogen (typ. 0-140)

Phosphorus (typ. 5-145)

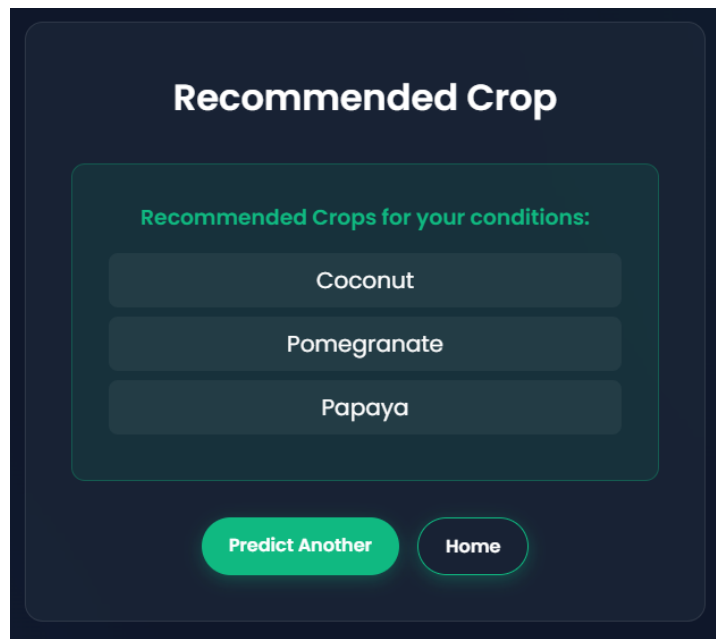
Potassium (typ. 5-205)

Temperature (°C) (typ. 5-50)

Humidity (%) (typ. 10-100)

pH Value (0-14) (typ. 3-10)

Rainfall (mm) (typ. 0-400)

A dark-themed web interface for "Crop Yield Prediction". The header includes the "AgriAI" logo and navigation links: Home, Dashboard, About, Services, and Contact. The main content area contains a form with eight input fields arranged in two columns: "Crop (e.g. Rice, Wheat)", "Crop Year (e.g. 2023)", "Annual Rainfall (mm/year)", "Pesticide (tonnes)", "Fertilizer (tonnes)", "Area (hectares)", "State (e.g. Assam)", and "Season (e.g. Kharif)". Each field has a placeholder text "Enter [field name]". Below the form is a blue "Predict Yield" button. Underneath the button is a light blue box labeled "Result" which displays "Predicted Yield: 2.49".

v. SUMMARY

This chapter explains the methodology behind the crop prediction and yield prediction models, including the steps taken for data processing, model training, and evaluation. The next chapter will discuss the technical details of the system's implementation and performance evaluation

B. Working Principle

i. Outline

This chapter details the technical aspects of the project, including coding solutions, system design, and performance analysis.

ii. Technical Coding and code solutions

The implementation Involves the following steps: -

Step 1: Data Preparation and Preprocessing

- Load Dataset: Import and clean the dataset (remove duplicates, unnecessary columns, and validate fields).
- Preprocessing:
 - Scale numerical columns using Standard Scaler.
 - Encode categorical columns using OneHotEncoder or Label Encoding.
 - Split the data into training (80%) and testing (20%) subsets.

Step 2: Model Training and Evaluation

- Train Model: Use appropriate machine learning algorithms (e.g., Random Forest Classifier for classification tasks or Decision Tree Regressor for regression tasks).
- Evaluate Model:
 - Use metrics like precision, recall, and F1-score for classification tasks.
 - Use Mean Absolute Error (MAE) and R-squared (R^2) for regression tasks.

Step 3: Web Application Deployment

- Save Models: Store trained models and preprocessing pipelines as .pkl files.
- Deploy Web App: Build a Flask web app to:
 - Accept user inputs, preprocess them, generate predictions, and display results

iii. Web Application Deployment

- **Crop Recommendation Form Purpose:** Collects soil and environmental data (e.g., Nitrogen, Phosphorus, etc.) to predict the best crop. Action: User submits numeric inputs; system processes and suggests a crop.
- **Crop Yield Prediction Form Purpose:** Gathers climatic, regional, and crop-related data to estimate crop yield. Action: User provides inputs (numeric and categorical); system predicts yield.
- **Result Display Page for Recommendation:** Displays the suggested crop based on user inputs. For Yield Prediction: Shows the estimated crop yield in hectograms per hectare.

iv. Prototype A prototype was developed with:

- **Frontend:** Developed using HTML, CSS, and JavaScript to create responsive and interactive user interfaces for input forms and result display.
- **Backend:** Built using Python Flask to handle input processing and machine learning model predictions.
- **Sample Data:** Simulated dataset for validation.

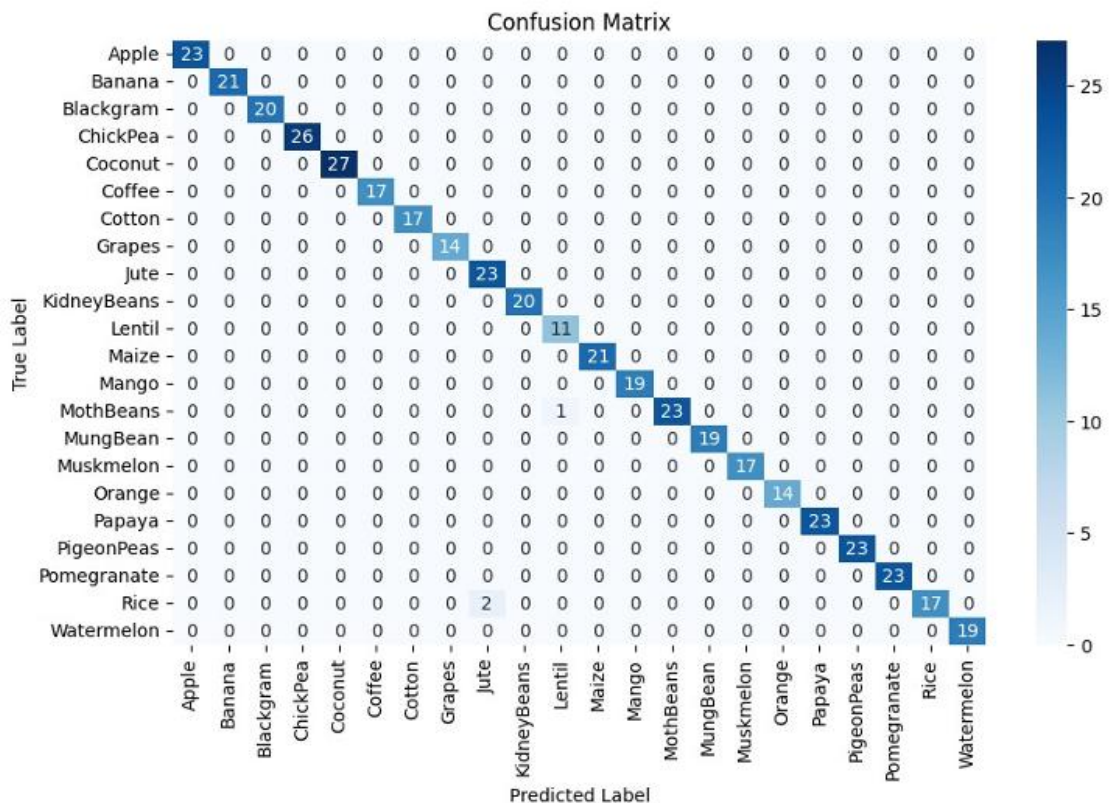
C. Results and Discussion

i. Crop Recommendation Model – Results

The Crop Recommendation model was evaluated using standard classification metrics such as Accuracy, Precision, Recall, and F1-Score.

The overall model accuracy achieved was 99.32%, indicating excellent classification capability across multiple crop categories.

➤ Confusion Matrix Analysis



The confusion matrix graph visually compares the actual crop labels with the predicted crop labels.

Discussion:

- Most values are concentrated along the diagonal of the matrix, indicating correct classifications.
- Crops such as Apple, Banana, Coconut, Mango, Papaya, and Pomegranate show perfect or near-perfect classification.
- Very few off-diagonal values are observed, indicating minimal misclassification.
- Slight confusion was observed in crops such as Rice (recall slightly lower), suggesting some overlap in environmental features with similar crops.

Overall, the confusion matrix confirms that the classification model performs extremely well across different crop classes.

➤ Classification Metrics Interpretation

The precision, recall, and F1-score values are close to 1.00 for most crop categories.

- High precision indicates that when a crop is predicted, it is almost always correct.
- High recall shows that most actual crop instances are correctly identified.
- High F1-score confirms balanced performance even in multiclass conditions.

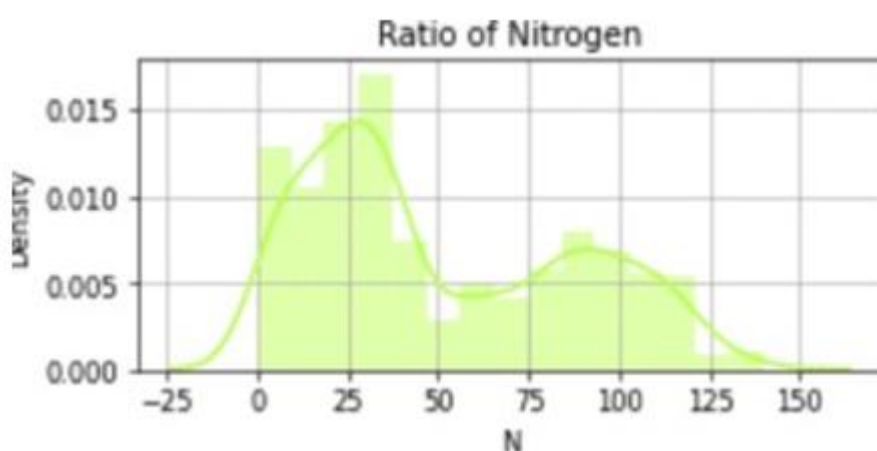
These metrics validate the robustness and stability of the crop recommendation model.

ii. Performance Analysis Using Feature Distribution Graphs

The soil and environmental parameters were analyzed using graphical distributions (Nitrogen, Phosphorus, Potassium, Temperature, Humidity, pH, and Rainfall)

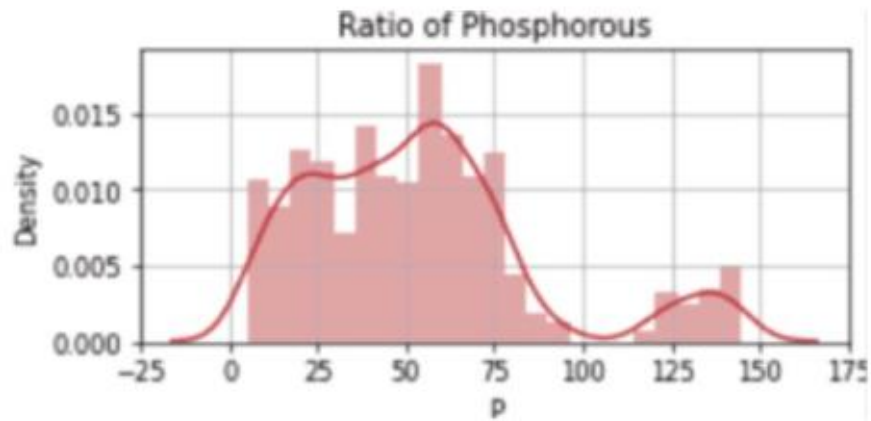
➤ Nitrogen Distribution

Nitrogen values follow an approximately normal distribution with a peak in the mid-range, indicating balanced nutrient levels across samples.



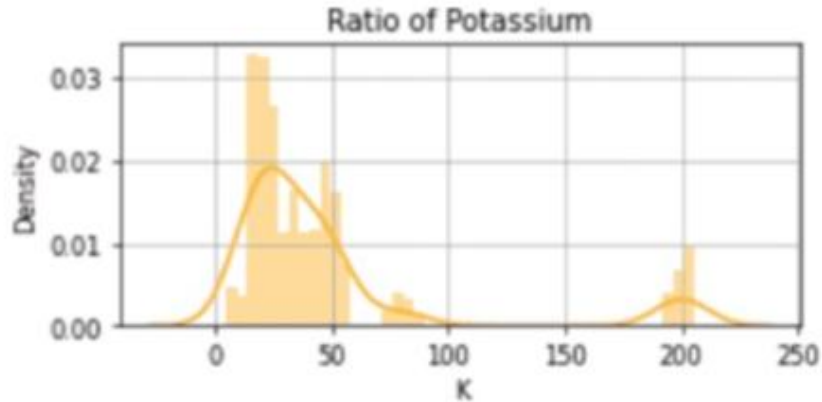
➤ **Phosphorus Distribution**

Phosphorus shows a multimodal distribution, suggesting variability across different agricultural regions



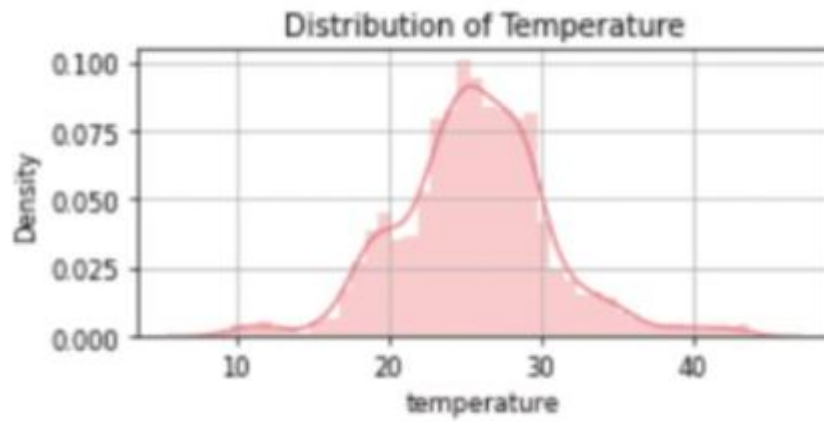
➤ **Potassium Distribution**

Potassium exhibits a bimodal pattern, indicating the presence of two dominant soil groups.



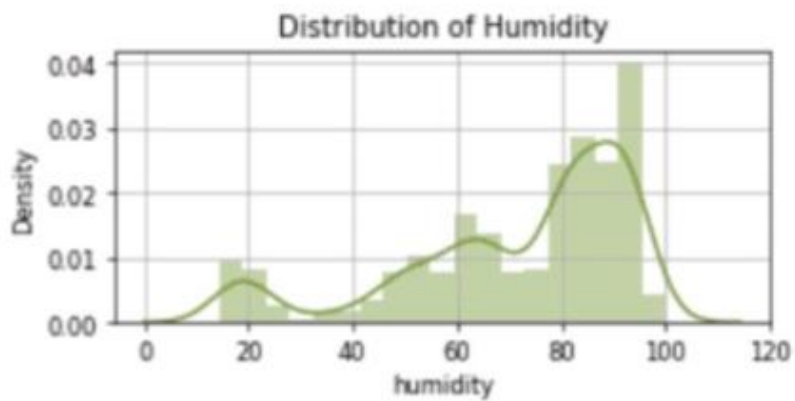
➤ **Temperature Distribution**

Temperature follows a bell-shaped curve, representing relatively stable climatic conditions.



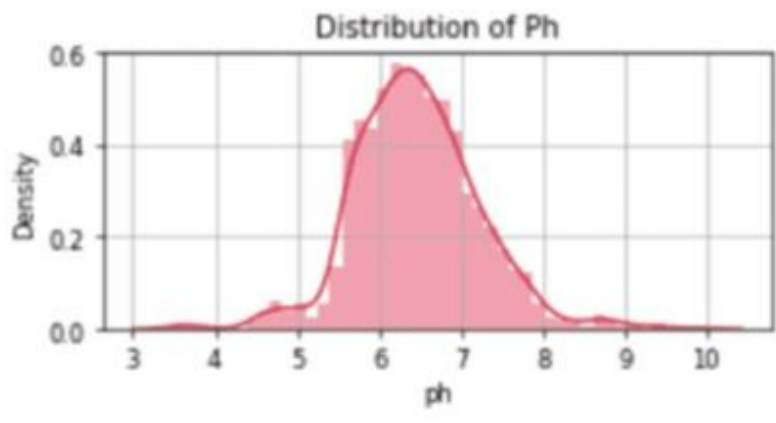
➤ **Humidity Distribution**

Humidity values are concentrated in higher percentage ranges, which supports crop growth modelling



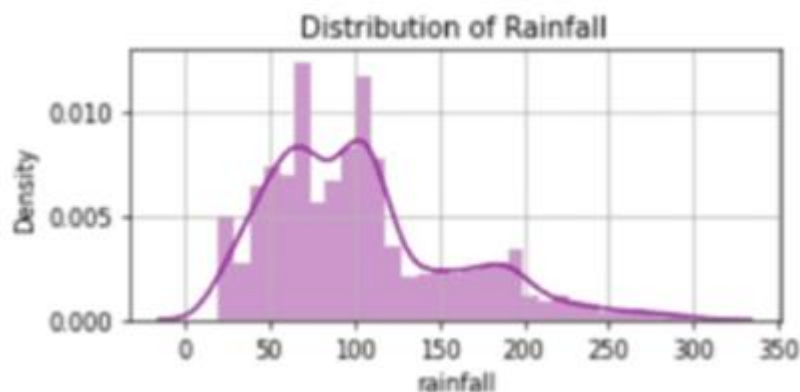
➤ **pH Distribution**

pH values are centered around 6, indicating slightly acidic soil conditions suitable for many crops.



➤ **Rainfall Distribution**

Rainfall distribution is right-skewed, meaning moderate rainfall is common while heavy rainfall is less frequent.



These feature distributions confirm that soil nutrients and climatic factors significantly influence crop suitability and model prediction accuracy.

iii. Yield Prediction Model Results

The Yield Prediction model was evaluated using regression metrics including Mean Absolute Error (MAE) and R-squared (R^2).

The Decision Tree Regressor achieved:

- MAE: 8.26
- R^2 Score: 0.97

- Overall prediction accuracy approximately 96–97%

➤ **Regression Performance Interpretation**

An MAE of 8.26 indicates that the predicted yield values are very close to the actual values. An R^2 score of 0.97 shows that 97% of the variance in crop yield is explained by the model, confirming strong predictive performance.

The Decision Tree model was selected due to its interpretability, computational efficiency, and suitability for agricultural datasets.

iv. Overall Discussion

The results demonstrate that the proposed system performs with high accuracy and reliability.

Key observations include:

- Very high classification accuracy (99.32%) in crop recommendation.
- Strong regression performance ($R^2 = 0.97$) in yield prediction.
- Minimal misclassification across crop types.
- Effective use of soil and environmental parameters.

The system successfully integrates soil nutrients, climatic conditions, and historical data to provide accurate, data-driven agricultural recommendations suitable for real-world applications.

D. Individual Contribution by Members

1. Divyansh Kumar (Registration Number: 23BAI10514)

Divyansh Kumar mainly worked on collecting and preparing data that formed the foundation of our entire project. Since our system depends heavily on accurate agricultural data, he ensured that all datasets related to soil nutrients, rainfall, temperature, humidity, pH values, and crop yield were collected from reliable sources. He spent significant time cleaning the data by removing duplicate entries, handling missing values, and correcting inconsistencies.

He carefully checked whether all required features were present and made sure the data was properly structured before it was used for model training. Divyansh also analyzed how different parameters like Nitrogen, Phosphorus, Potassium, and rainfall were distributed in the dataset, which helped us better understand the data before building the models. His efforts ensured that our machine learning models were trained on clean and meaningful data, reducing errors and improving reliability. He also helped in documenting the preprocessing steps clearly in the report.

He also integrated the location API, which is integrated to National Agricultural dataset.

2. Aayush Singh (Registration Number: 23BCE10279)

Aayush Singh was responsible for building the Crop Recommendation system. He worked on implementing the Random Forest Classifier after comparing it with other machine learning algorithms. His role included defining the input features such as soil nutrients and environmental conditions and setting the correct target variable for classification.

He divided the dataset into training and testing sets to properly evaluate the model's performance. After training the model, he analyzed its accuracy using metrics like precision, recall, F1-score, and confusion matrix. He also studied feature importance to understand which soil and climatic factors influenced the crop prediction the most.

Aayush ensured that the trained model performed accurately and consistently before saving it for integration into the web application. His contribution was central to achieving high prediction accuracy and making the crop recommendation system dependable and practical.

3. Amogh Agarwal (Registration Number: 23BCE10220)

Amogh Agarwal worked on developing the Yield Prediction model of the project. He implemented the Decision Tree Regressor to estimate crop yield based on inputs such as rainfall, temperature, pesticide usage, region, and crop type. Before training the model, he carefully preprocessed the data by scaling numerical values and encoding categorical variables.

He used tools like StandardScaler and OneHotEncoder to ensure that the data was properly formatted for regression analysis. Amogh trained and tested the model using proper data splitting and evaluated its performance using Mean Absolute Error (MAE) and R-squared (R^2) values.

He also compared multiple regression algorithms before selecting the most suitable one for our system. His work ensured that the yield predictions were both accurate and computationally efficient. He also contributed to documenting the regression process and explaining the results in the report.

4. Roshmik Agrawal (Registration Number: 23BAI10014)

Roshmik Agrawal was responsible for designing and developing the frontend of our web application. His main goal was to make the system simple, clean, and easy to use so that even non-technical users, especially farmers, could interact with it comfortably. He used HTML, CSS, and JavaScript to design structured input forms for both the Crop Recommendation and Yield Prediction modules.

He carefully planned the layout so that users could clearly understand what data they need to enter, such as soil nutrients, rainfall, temperature, and other parameters. He implemented proper input validation to prevent incorrect or incomplete entries, ensuring smoother functionality. Roshmik also designed the result display page in a way that predictions are shown clearly and understandably without technical complexity.

He focused on improving user experience by maintaining consistency in fonts, spacing, and overall layout. He tested the interface multiple times to ensure it worked smoothly across different devices and browsers. His contribution played a major role in making the project visually appealing, user-friendly, and practical for real-world use. He also assisted in taking screenshots and preparing frontend-related explanations for the project report and presentation.

5. Mohd. Belal (Registration Number: 23BAI10866)

Mohd. Belal handled the backend development and full system integration of the project. He used the Flask framework to connect our trained machine learning models with the frontend interface, ensuring smooth communication between user input and prediction output. His work transformed our separate modules into a complete working web application.

Belal implemented backend routes for both crop recommendation and yield prediction. He ensured that when users submit input data, it is properly validated, formatted, and passed to the respective trained model for prediction. He also handled loading of saved model files and preprocessing pipelines to maintain consistency in predictions.

To ensure system stability, he added error-handling mechanisms to manage unexpected inputs or system failures. He tested the backend with multiple input combinations to verify accuracy and performance. Belal also worked closely with the frontend developer to resolve integration issues and improve system response time. His contribution was essential in making the project functional, stable, and ready for real-world deployment.

6. Surendra Mewada (Registration Number: 23MIP10136)

Surendra Mewada contributed significantly to system testing, validation, and overall quality assurance of the project. After the integration of frontend, backend, and machine learning models, he focused on verifying that the entire system worked correctly and consistently. He conducted multiple functional tests using different combinations of input values to ensure reliable and logical predictions.

Whenever errors or mismatches were identified during testing, Surendra assisted in debugging and resolving integration issues. He carefully reviewed the outputs of both the crop recommendation and yield prediction modules to confirm their accuracy. He also cross-checked evaluation metrics such as confusion matrix, classification report, Mean Absolute Error (MAE), and R^2 values to ensure consistency in performance results.

In addition to technical validation, Surendra supported report formatting, proofreading, and final compilation. He ensured that the documentation was clear, organized, and properly structured. His contribution strengthened the reliability, stability, and overall quality of the final project before submission and presentation.

4. CONCLUSION

The project demonstrates the potential of integrating Machine Learning with GPS-based location intelligence to enhance agricultural decision-making. By combining soil parameters, climatic conditions, and real-time geographic data, the system provides scientifically grounded crop recommendations and yield predictions. The strong performance of the models, validated through metrics such as confusion matrix analysis, classification report, R^2 score, and Mean Absolute Error, confirms the reliability and robustness of the approach. High classification accuracy and minimal misclassification indicate that the system can effectively guide farmers in selecting suitable crops and estimating expected production, thereby improving productivity and reducing financial risks.

Despite its promising results, the system is subject to certain limitations. Its performance heavily depends on the availability and quality of agricultural datasets, including soil properties, climatic records, and yield data. Incomplete or inaccurate data can reduce prediction accuracy and affect reliability. Additionally, models trained on region-specific datasets may not generalize perfectly to other geographical areas with different soil compositions, climatic conditions, or farming practices. Environmental factors such as temperature and rainfall are dynamic and continuously evolving, requiring periodic updates to maintain model accuracy. Furthermore, limited access to digital infrastructure in rural areas may restrict adoption among small-scale farmers. Language barriers and usability concerns can also affect accessibility for non-technical users.

To address these limitations and enhance system capabilities, future improvements can focus on integrating real-time IoT-based soil sensors, advanced weather APIs, and satellite imagery to enable continuous environmental monitoring. Expanding multilingual support and localizing the interface to regional farming practices can significantly improve usability and adoption. The inclusion of pest and disease prediction modules would further strengthen the advisory capabilities of the system. Collaboration with agricultural agencies and research institutions can help expand datasets for better accuracy and broader applicability. Developing cloud-based and mobile-accessible platforms will improve scalability and outreach, especially in remote areas. Additionally, extending the system to provide personalized fertilizer and irrigation recommendations can further optimize resource utilization. Continuous refinement of machine learning models with updated data and advanced algorithms will ensure long-term scalability and performance.

Overall, the proposed system represents a significant step toward modernizing agriculture through intelligent, data-driven solutions. By leveraging machine learning, geo-spatial intelligence, and web-based deployment, the platform supports informed decision-making, promotes sustainable farming practices, and enhances farmer profitability. With continued refinement and integration of advanced technologies, this system has the potential to become a comprehensive digital advisory tool capable of transforming precision agriculture and contributing to long-term agricultural sustainability.

5. References:

- [1] A. Priyadharshini, S. Chakraborty, A. Kumar, and O. R. Pooniwala, “Intelligent Crop Recommendation System Using Machine Learning,” *IEEE Xplore*, 2021.
- [2] S. A. Z. Rahman, K. C. Mitra, and S. M. M. Islam, “Soil Classification Using Machine Learning Methods and Crop Suggestion Based on Soil Series,” in *Proc. ICCIT*, 2018.
- [3] N. Singh, D. Pant, D. P. Singh, and B. Pant, “Crop Prediction Method to Maximize Crop Yield Using Machine Learning,” 2020.
- [4] R. Surana and R. Khandelwal, “Crop Yield Prediction Using Machine Learning: A Pragmatic Approach,” 2022.
- [5] M. Kamilaris and F. X. Prenafeta-Boldú, “Crop Yield Prediction Using Machine Learning: A Systematic Literature Review,” *Computers and Electronics in Agriculture*, vol. 151, pp. 61–69, 2020.
- [6] A. Ghosh et al., “Agricultural Recommendation System for Crop Protection,” *Computers and Electronics in Agriculture*, vol. 159, pp. 221–230, 2018.
- [7] X. Lin, “Use of Machine Learning Techniques in Soil Classification,” *Sustainability*, vol. 15, no. 3, 2023.
- [8] R. Patel and S. Kumar, “Precision Agriculture with IoT: Real-Time Crop Monitoring,” *International Journal of Smart Farming*, vol. 8, no. 2, 2023.
- [9] A. Gupta and P. Sharma, “Soil Nutrient Analysis for Crop Suitability Using Machine Learning,” *Agricultural Data Science*, vol. 12, no. 4, 2021.
- [10] M. Johnson and T. Brown, “Weather-Based Crop Yield Prediction Using Machine Learning,” *Journal of Climate and Agriculture*, vol. 10, no. 1, 2020.
- [11] S. Vaishnavi et al., “Agricultural Crop Recommendations Based on Productivity and Season,” 2021.
- [12] S. K. Apat, J. Mishra, K. S. Raju, and N. Padhy, “AI-Based Crop Recommendation System Using Machine Learning,” *Journal of Scientific & Industrial Research*, vol. 82, 2023.
- [13] T. G. Keys, “Predicting USCS Soil Classification Using Random Forest,” *Journal of Soil Mechanics*, 2016.
- [14] J. Smith and K. Lee, “AI-Based Crop Recommendation System,” *Journal of Agricultural Technology*, vol. 15, no. 3, 2022.
- [15] R. Singh and V. Yadav, “Fertilizer Recommendation System Based on Soil Conditions,” *Journal of Sustainable Agriculture*, vol. 14, no. 2, 2022.

- [16] S. Ramesh and R. Bharathi, “Machine Learning in Agriculture: A Survey,” *IEEE Access*, vol. 7, pp. 160–170, 2019.
- [17] Y. Liakos et al., “Machine Learning in Agriculture: A Review,” *Sensors*, vol. 18, no. 8, 2018.
- [18] A. Wolfert, L. Ge, C. Verdouw, and M. Bogaardt, “Big Data in Smart Farming,” *Agricultural Systems*, vol. 153, 2017.
- [19] J. L. Hatfield and J. H. Prueger, “Temperature Extremes and Crop Production,” *Weather and Climate Extremes*, vol. 10, 2015.
- [20] FAO, “Climate Change and Food Security,” Food and Agriculture Organization, 2020.
- [21] OpenWeatherMap API Documentation, 2024.
- [22] HTML5 Geolocation API Specification, W3C Recommendation.
- [23] L. Breiman, “Random Forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [24] J. R. Quinlan, “Induction of Decision Trees,” *Machine Learning*, vol. 1, no. 1, pp. 81–106, 1986.
- [25] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, Springer, 2009.
- [26] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*, MIT Press, 2016.
- [27] Scikit-learn Documentation, “Random Forest Classifier,” 2024.
- [28] Scikit-learn Documentation, “Decision Tree Regressor,” 2024.
- [29] Flask Documentation, Pallets Projects, 2024.
- [30] Kaggle Dataset: Crop Recommendation Dataset, 2023.

6. Biodata with Picture:

1. Divyansh Kumar

- Registration Number: 23BAI10514
- Email: divyansh.23bai10514@vitbhopal.ac.in
- Education: VIT Bhopal University – CGPA – 8.47
- Skills:
 - Python, C, C++, HTML, CSS, Javascript, SQL, Java
 - Machine Learning, Web Development, Content Writing, AWS Services
 - Database, Dynamo DB, Mongo DB



2. Aayush Singh

- Registration Number: 23BCE10279
- Email: aayush.23bce10279@vitbhopal.ac.in
- Education: VIT Bhopal University – CGPA – 8.39
- Skills:
 - Programming: C, C++, Java, Python, JavaScript
 - Web Technologies: HTML5, CSS3, JavaScript, React.js
 - Tools & Platforms: Git, GitHub, Postman
 - Core Concepts: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks



3. Amogh Agarwal

- Registration Number: 23BCE10220
- Email: amogh.23bce10220@vitbhupal.ac.in
- Education: VIT Bhopal University – CGPA – 8.45
- Skills:
 - Programming: C++, Java, Python, JavaScript
 - Web Technologies: HTML, CSS, JavaScript
 - Tools & Platforms: Git, GitHub, Postman
 - Core Concepts: Data Structures & Algorithms, OOPS, Operating Systems, Computer Networks



4. Roshmik Agrawal

- Registration Number: 23BAI10014
- Email: roshmik.23bai10014@vitbhupal.ac.in
- Education: VIT Bhopal University – CGPA – 8.44
- Skills:
 - Programming: Java, C++, Python, C
 - Web Technologies: HTML, CSS, JavaScript
 - Tools & Platforms: Git, GitHub, MATLAB
 - Core Concepts: Data Structures & Algorithms, OOPS, Operating Systems, DBMS, Computer Networks



5. Mohd. Belal

Registration Number: 23BAI10866

Email: mohd.23bai10866@vitbhopal.ac.in

Education: VIT Bhopal University – CGPA – 7.34

Skills:

- Programming: Java, C++, Python, C
- Web Technologies: HTML, CSS, JavaScript
- Computer Vision: OpenCV, YOLOv8, Image Processing
- Frameworks & Libraries: Tensor Flow, Scikit-Learn, NumPy, Pandas, Matplotlib
- Tools & Platforms: Git, GitHub,
- Core Concepts: Data Structures & Algorithms, OOPS, Operating Systems, DBMS,
- Computer Networks



6. Surendra Mewada

Registration Number: 23MIP10136

Email: surendra.23mip10136@vitbhopal.ac.in

Education: VIT Bhopal - CGPA – 7.0

Skills:

- Python, C++
- Pandas, NumPy
- Matplotlib
- MySQL,
- Jupyter Notebook, VS Code, Pycharm

